

Private and Public Sector Managers: An Empirical Investigation of Job Characteristics and Organizational Climate

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This empirical research compared job characteristics and organizational climate in the private and public sectors. Perceptions and satisfaction of 240 top managers from a variety of private and public organizations in Israel were compared. The hypotheses tested were that (a) performance-based rewards and (b) policies that promote efficiency would be significantly more prevalent in private sector organizations and (c) that higher levels of satisfaction would be expressed by managers in the private sector. Two-way analyses of variance, private versus public and production versus service organizations, were performed. Results yielded significant main effects for sector of ownership and provided strong support for the three hypotheses of the research. Interaction effects in specific scales were also in the hypothesized direction, revealing that the private/public sector differences are further amplified in service organizations. The relation of the findings of this research to those of comparable studies in the United States are discussed.

In recent years there has been an increasing interest in research comparing the private and public sectors, with a view toward improving productivity in the public sector. Despite the apparent importance of the topic and the numerous dimensions on which private and public organizations can be compared (Rainey, Backoff, & Levine, 1976), few empirical studies have appeared in the literature (for example, Buchanan, 1974, 1975; Paine, Carroll, & Leete, 1966; Porter & Mitchell, 1967; Rawls, Ulrich, & Nelson, 1975; Schuster, Colletti, & Knowles, 1973). This research survey directly compared practices in the private and public sectors based on responses of top managers from a broad spectrum of Israeli organizations in each sector.

The private—public distinction is not a clear-cut dichotomy. Walmsley and Zald (1973) base their differentiation on patterns of ownership and funding. However, some government organizations are financed by user charges, and certain private organizations are primarily reliant on government contracts. Authors like Dahl and Lindblom (1953) and Fottler (1981) have proposed a continuum between private and public sector organizations. Fottler identified four classes of organizations within the continuum between profit-making and governmental agencies. Nevertheless, most large business firms and large government agencies in the United States and in Israel fall clearly into the private or public domain.

Management functions, whether in the private or the public sector, are conceptually similar. However, the contentions by the generic school of management, that management functions are the same in all types of organizations, have been challenged (Fottler, 1981; Rainey et al., 1976).

Following Thompson's (1962) notion, effectiveness depends on matching the internal organizational structure to the demands of the task environment; it would therefore appear that effective

functioning of the private and public organizations would depend on different criteria, because each must adapt to different environmental contingencies. For example, one would expect differences between the two sectors in the sources of funding and ownership, the legal and political environments, and the criteria for accountability and control, to have a direct impact on the structure and practices in the two types of organizations.

Among the environmental factors most often cited in the literature as differentially affecting organizations in the two sectors, are differences in the degree of market exposure and the sources of funding (Lindblom, 1977; Turk, Walmsley, & Zald, 1979; Walmsley and Zald, 1973). Managers in the private sector operate in a competitive and dynamic environment, where profitability is the ultimate criterion of success. In order to achieve these objectives, they are expected to be responsive to market and customer demands and to implement organizational policies that facilitate efficient and economically rational action. The private sector economic dynamics are defined as a dependence on the market for resources, which provide incentives and automatic penalties that reinforce policies for cost reduction, operating efficiency, and effective performance.

Managers in the public sector are required to maintain constituencies, seek multiple goals, and obtain funding through an appropriation process, which is susceptible to political influences (Porter & Van Maanen, 1970; Rainey et al., 1976). The culture of the public sector is one of conflicting values, where the special restrictions and limitations posed by the competing objectives interfere with the actual attainment of productivity and efficiency goals (Von Mises, 1944). The common practice of obtaining resources through an appropriation process creates conditions in which political considerations take priority and reduce the relative emphasis on operational and allocational efficiency (Drucker, 1973; Levine, 1969; Niskanen, 1971; Savas, 1974; Schultze, 1970; Rainey, 1979). Because public sector managers are also expected to meet demands of public responsiveness, they are faced with constant conflict between the goals of accountability and efficiency (Whorton & Worthley, 1981).

In the private sector, the extent to which certain actions will

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be encouraged and rewarded is generally determined by the degree to which they help meet the goals of increased efficiency. Conversely, the public sector, facing multiple and competing goals and additional constraints, might assign a lower priority to policies that reward efficiency, yielding more to the vagaries of the political climate.

These considerations led to the first hypothesis: In broad terms, the more an organization's survival depends on functioning within a competitive market, the more it will emphasize and reward efficient performance.

Hypothesis 1. Emphasis on use of performance-based rewards will be significantly more prevalent in the private sector compared to public sector organizations, as reported by their respective managers.

Rewards are just one component of management practices. Other practices of work organizations that influence the perceptions and behavior of employees have been studied in the literature on job design (Hackman & Lawler, 1971; Hackman & Oldham, 1976; Turner & Lawrence, 1965) and in organizational climate research (Likert, 1967; Litwin & Stringer, 1968; Campbell, Dunnette, Lawler, and Weick, 1970; Schneider, 1972, 1981; Payne & Pugh, 1976). Certain of these practices directly relate to the organization's efficiency, others do not. Management practices that address improvement of methods and efficiency, task autonomy and significance, reward systems, and task clarity are of direct relevance to efficiency; others, such as participation, openness, and task variety, are not directly efficiency-related. A second, more general hypothesis on the differences between private and public sectors was therefore advanced.

Stated in broad terms, differences between practices of the private and public sectors are expected to be primarily in efficiency-related areas cited above, because the major differences between the two sectors lie in their respective sources of funding, and the consequent productivity and efficiency requirements. Private sector managers are therefore expected to view their organizations as more actively pursuing policies that address efficiency, whereas the differences between the two sectors will not be significant in the nonefficiency-related areas. This is expressed in the second hypothesis:

Hypothesis 2. Policies that promote efficiency will be significantly more prevalent in private sector organizations compared to public sector organizations, as reported by their respective managers.

Previous research comparing behavior in the private and public sectors reveals differences in levels of satisfaction between the two sectors. Paine, et al. (1966), Porter and Mitchell (1967), Rhinehart, Barrell, DeWolfe, Griffin, and Spaner (1969), and Buchanan (1974) report lower levels of need satisfaction with work and lower levels of organizational commitment on the part of government employees.

Differences in satisfaction between the private and public sectors are expected to be associated with differences between the two sectors in conditions that affect motivation and satisfaction. Motivation theories attribute the origins of satisfaction to specific conditions in the work environment. Literature on achievement motivation (Atkinson, 1958; McClelland, 1961), on goal setting (Locke, 1968), and the expectancy theories of motivation (Lawler, 1971; Porter & Lawler, 1968; Vroom, 1964) suggest that challenge, clear goals, feedback, and rewards contingent on perfor-

mance contribute to positive motivational states and satisfaction. Literature on task design points to the "core job dimensions" of task autonomy, variety, identity, significance, and feedback (Hackman & Lawler, 1971; Hackman & Oldham, 1976) as associated with satisfaction, particularly among specific populations. Conversely, research on role stress (see Van Sell, Brief, & Schuler, 1980, for a review), points to the deficiencies in role clarity (i.e. role ambiguity) as a major source of stress and dissatisfaction in organizations. With specific reference to the public sector, Buchanan (1975) suggests that a deficiency in goal clarity, which characterizes the multiple, conflicting, and often intangible objectives typical of the public sector, contributes to lower satisfaction and commitment levels. He concludes that the contradictions and frequent diffuseness of public organizational goals interfere with the individual's perception of his or her own role, and of the impact of his or her behavior on overall organizational efficiency.

In this study, perceptions and satisfaction with job and organizational climate were independently assessed. Because differences between the private and public sectors in the perceived presence of these characteristics were predicted, differences between the two sectors in satisfaction with these characteristics are also expected. This led to our third hypothesis:

Hypothesis 3. Satisfaction with job and organizational climate characteristics will be significantly higher among private sector compared to public sector managers.

Method

Sample

Two hundred and forty top managers from a broad range of Israeli work organizations participated in the study. Half the sample consisted of managers from the private sector, the other half of managers from the public sector. Table 1 presents the breakdown of the sample. As indicated in this table, there was an additional dichotomy to the sample: Sixty of the 120 private sector managers, and 60 of the 120 public sector managers were employed in production organizations. The remaining 60 of each sector were employed in service organizations.

Production organizations in both sectors represented the following industries: metals and electronics, textile, clothing and leather, food, chemicals and plastic, and various others. Production organizations of the public sector in this study refer to industries owned by the Histadrut, the general Federation of Labor. Service organizations included financial institutions, banks, insurance agencies, and others, as well as government organizations such as the Education, Finance, and Interior Ministries. The sampling of the organizations was a stratified random one, and was roughly proportional to the profile of industrial branches in Israel as appearing in the Statistical Abstracts of Israel (1977), Shaliv (1977), and the Homogeneous Classification of Economic Branches (1973).

Individuals in the three top levels of management were included in the sample. Of all questionnaires handed out, 68% were completed and used in the sample. Eighty percent of these questionnaires were personally administered; the remainder were returned by mail, resulting in a total of 240 completed questionnaires.

The median age of respondents in the study was 44 years, and in 49.6% of the cases the managers had been employed by their companies for over 10 years. The managers held their present job for 3 to 7 years in 43.3% of the cases, whereas 30.0% of them had been on the same job for over 8 years. The categories of their functional specialization included Marketing and Finance (29.9%), Personnel (9.7%), Research and Engineering (7.2%), and Production (3.3%). Eleven percent of the respondents viewed

Table 1
Composition of the Sample

Type of organization	Sector of ownership			Total
	Private sector	Public sector	Histradrut	
Production organizations (metals & electronics, textile, clothing & leather goods, chemicals & plastic, & other)				
N	60	—	60	120
Service organizations (financial institutions, banks, insurance agencies, others, government organizations, Ministries of Education, Finance, Interior, other)				
N	60	60	—	120
Total (N)	120	120	120	240

themselves as performing more than one function, and 38.6% belonged to other classifications, including General Management.

Two-way analyses of variance (private/public and production/service) were performed on the background variables of the research, to examine the differences among samples and subsamples. Out of 17 background variables examined (including the manager's level of education, area of study, age, tenure with the job and with the organization, functional specialization, country of origin, etc.) only four yielded significant differences. Private sector managers had significant higher mean income than public sector managers, as expected, and the education level of public service employees was higher than any other group, because rigorous entrance examinations are the criterion for hiring public sector employees at all levels. Finally, age and tenure with the organization revealed interaction effects: Private service and public production managers had a significantly higher mean age and tenure with the organization than did the other two groups.

Measures

Two questionnaires were used for the collection of the data for this research, one addressing job characteristics, the other organizational climate. Both questionnaires had been developed by Solomon (1980, 1985) using the facet analytic approach (Guttman, 1954, 1959). Facet analysis was applied in defining the constructs, Smallest Space Analysis (SSAI; Guttman, 1968; Lingoes, 1973) was used for the empirical testing of the structural relations among the questionnaire variables.

The job characteristics questionnaire consisted of 20 items. These included job reward policies, interpersonal job aspects, and training opportunities. Other dimensions such as task autonomy, significance, variety, and feedback corresponded to Hackman and Lawler's (1971) and Hackman and Oldham's (1976) core job dimensions and, finally, a group of items referred to role clarity-ambiguity studied in the literature on role stress (Kahn, Wolfe, Quinn, Snoek, & Rosenthal, 1964; House & Rizzo, 1972; Rizzo, House, & Lirtzman, 1970).

The Organizational Climate Questionnaire consisted of 19 items. These pertained to policies of improvement of methods and efficiency, partic-

ipation, reward structure, consideration, and decision making practices (Campbell et al., 1970; Likert, 1967; Payne & Pugh, 1976).

Two independent assessments were made of each of the 20 job characteristics and each of the 19 organizational climate items: The two assessments were as follows: (a) perceived existence of these characteristics and (b) satisfaction reported with each one of them.

Perceived existence of each job and organizational climate characteristic was measured by asking the respondents to indicate the extent to which each item was present. The range consisted of six response categories from "to a very little extent" up "to a very great extent."

Satisfaction responses with the various job and organizational climate attributes were assessed by asking the individual respondents to choose among six alternatives ranging from "not at all satisfied" to "very satisfied."

Results

Factor Analysis

Responses from private and public sector managers were combined and were subjected to a principal axis factor analysis with a varimax rotation. Items from both questionnaires, job characteristics and organizational climate, were jointly factor analyzed in order to identify the number and composition of the factors. The joint factor analysis aimed at testing the empirical independence of the two questionnaires. Table 2 presents the results of the factor analysis for the existence responses.

For each questionnaire item indicated on the left, the factor loadings on each factor are depicted. Eigenvalues and percentage of total variance accounted for are also presented for each factor. The factor analysis of the existence responses (see Table 2) resulted in six factors with eigenvalues greater than 1.00, which explained 89% of the total variance. The Cronbach alphas for the six factors ranged from .68 to .81.

As the factor analyses of both assessments indicate, there was very little commonality between the two questionnaires. The only common items were three job characteristics items, and one organizational climate item. The other 35 items loaded on distinctly unique factors, one set for job characteristics items and one for organizational climate items. Three of the six factors of the perceived existence factor analysis clearly pertained to job characteristics (Factors 1, 5, and 6), whereas the remaining three factors (2, 3, and 4) only included organizational climate items. In the satisfaction assessment, job characteristics items loaded on two factors (1 and 4) and the organizational climate on the remaining three (2, 3, and 5).

Three of the six factors that emerged from the existence assessment factor analysis (Table 2) pertained to job characteristics. Factor 1, which was labeled Task Autonomy and Significance, included five job characteristics items. Four of these corresponded to three of Hackman and Oldham's core job dimensions. Sample item: "Independence to make important decisions and determine work objectives." Factor 5, referred to as Task Clarity, included seven job characteristics items and one organizational climate item. Sample item: "Clearly defined authority and areas of responsibility." Finally, Factor 6 referred to as Task Variety and Development, included four job characteristics items. Sample item: "A job that demands continued skill development." Cronbach alpha (Nunnally, 1967) estimates of internal consistency reliability for Factors 1, 5, and 6 treated as scales were .81, .77, and .68, respectively.

The remaining three factors of the existence assessment factor analysis corresponded to items of the organizational climate questionnaire. Factor 2, which was dominated by items pertaining to Improvement of Methods and Efficiency, consisted of five organizational climate items. Sample item: "Resources, for systematic training and improvement." Factor 3, which contained items on Participation and Openness, included eight organizational climate items and one job characteristics item. Sample item: "Teamwork and cooperation at all levels in the organization in seeking common objectives." Factor 4, which pertained to the organization's Reward Orientation, included five organizational climate items and two job characteristics items. Sample item: "Monetary reward system contingent on employee contribution

to the organization." One item, number 14, is not presented in these tables, because it did not load highly on any factor and is not included in the other data analysis. Cronbach alpha estimates of internal consistency reliability for Factors 2, 3, and 4 treated as scales were .81, .77, and .81, respectively.

The private/public sector comparisons of the existence responses relied on the above six factors of the existence assessment, treated as scales. For purposes of comparability, satisfaction responses were also examined by using the same scales based on the six factors of the existence factor analysis rather than the five factors of the satisfaction analysis. The existence factors were used to construct the scales because existence (perception) is a precursor to satisfaction. The Cronbach alpha estimates of re-

Table 2
Factor Analysis of the Questionnaires (Varimax Rotated Factor Matrix—Existence Assessment)

Item	Content	Factor 1: Task Autonomy and Significance	Factor 2: Improvement of Methods and Efficiency	Factor 3: Participation and Openness	Factor 4: Reward Orientation	Factor 5: Task Clarity	Factor 6: Task Variety and Development
Existence job characteristics							
1	Precise job definition	0.124	0.127	0.064	0.084	0.521	-0.027
2	Clear procedures	-0.057	0.028	0.044	0.189	0.630	0.057
3	Required interaction	0.082	0.017	0.108	-0.108	0.315	0.246
4	Friendship opportunities	-0.077	0.071	0.400	0.038	0.137	0.171
5	Resources for training	0.008	0.301	0.065	0.079	0.383	0.252
6	Complete tasks	0.560	0.025	0.178	0.109	0.262	0.010
7	Freedom to discuss problems	0.501	0.122	0.238	0.034	0.181	0.027
8	Opportunities for interaction	-0.053	-0.022	0.263	-0.062	0.201	0.521
9	Independence in decision making	0.795	0.137	0.145	0.108	0.052	0.203
10	Criteria for performance	0.210	0.144	0.103	0.250	0.652	0.175
11	Variety in job activities	0.379	0.022	0.159	0.107	-0.024	0.483
12	Monetary rewards acc. to contribution	0.296	0.002	0.026	0.690	0.153	0.067
13	Improvement of methods	0.026	0.319	0.067	0.339	0.316	0.174
15	Continued skill development	0.244	0.132	0.003	0.239	0.130	0.575
16	Opportunities for skill utilization	0.240	0.177	0.104	0.069	0.051	0.467
17	Clear authority and responsibility	0.460	0.185	0.050	0.094	0.589	-0.036
18	Feedback on performance	0.330	0.214	0.025	0.298	0.528	0.106
19	Independence of thought and action	0.677	-0.052	0.265	0.127	-0.072	0.115
20	Important & significant objectives	0.680	0.144	-0.019	0.210	0.170	0.318
Existence organizational climate							
1	Clear goals and objectives	0.088	0.358	0.389	0.162	0.386	0.021
2	Participative decision making	0.273	0.104	0.393	0.025	0.117	0.155
3	Acceptance and facing of problems	0.191	0.198	0.518	0.254	0.265	-0.077
4	Monetary rewards contingent	0.094	-0.035	0.054	0.852	0.144	0.151
5	Performance evaluation and feedback	0.117	0.224	0.213	0.456	0.179	-0.118
6	Improvement of working conditions	0.054	0.483	0.110	0.460	0.228	0.222
7	Training and efficiency improvement	0.100	0.653	0.067	0.027	0.096	0.165
8	Consideration for lower levels	0.113	0.514	0.440	0.225	0.034	-0.012
9	Communication and information flow	0.110	0.683	0.359	0.105	0.136	0.043
10	Teamwork and cooperation	0.244	0.418	0.475	0.155	0.027	-0.062
11	Openness in relations	0.194	0.196	0.549	0.125	-0.003	0.211
12	Reward for ability and leadership	0.182	0.028	0.414	0.434	0.149	0.179
13	Activities for employee welfare	-0.012	0.421	0.267	0.501	0.098	0.083
14	Emphasis on quality performance	0.234	0.405	0.260	0.418	0.127	-0.169
15	Improvement of work methods	0.012	0.597	0.294	0.104	0.232	0.046
16	Consideration for preferences	0.299	0.285	0.516	0.087	0.032	0.148
17	Rules and procedures	-0.128	0.376	0.058	-0.037	0.414	0.153
18	Freedom & personal responsibility	0.362	0.083	0.515	-0.037	0.170	0.024
19	Flexibility in decision making	0.305	-0.009	0.388	0.204	0.206	0.176
Eigenvalues		9.68	2.41	2.10	1.64	1.18	1.00
% of total variance accounted for		47.00	12.00	10.1	8.30	6.60	5.50

liability for the satisfaction responses on the six scales ranged from .72 to .82.

The existence scores and the satisfaction scores for the six factors were correlated, in order to examine the level of convergent validity between these two sets of measures of the same factors. The correlations between the existence and satisfaction scores along the convergent diagonal of the multitrait-multimethod matrix ranged from .74 to .88, indicating a high level of convergent validity (table available upon request).

Comparison of the Private and Public Sectors

Means and standard deviations were calculated for the existence and satisfaction assessments of the six scales. Consistently, existence and satisfaction means were higher in the private sector (tables available upon request), the differences between the two types of organizations being greatest in the satisfaction assessment.

To test for the significance of the differences between the mean responses from managers of the private and public sectors, two-way ANOVAs were conducted. Because half the sample of managers from each sector belonged to production and half to service organizations (Table 1), it was of interest to examine the effects of that dichotomy on their perceptions and levels of satisfaction. The two-way ANOVAs examined the effects of the private/public sector of ownership, and of the production/service dichotomy. Scheffé's contrasts were used in the analysis of the interactions. Tables 3 and 4 report the results of the two-way ANOVA for the different scales on the existence and the satisfaction assessments, respectively. The results of these analyses are presented by hypotheses.

Hypothesis 1

The results of the study provided strong support for Hypothesis 1, predicting significantly greater emphasis on performance-based rewards in the private sector. The two-way ANOVA of the Reward Orientation Scale (Table 3, existence assessment) produced highly significant main effects for sector of ownership, $F(1, 236) = 25.194, p < .000$. Private sector managers perceived that their rewards were contingent on performance to a much higher degree than public sector managers did (Private mean = 3.522, $SD = 0.86$, Public mean = 2.985, $SD = 0.82$). There was also a highly significant main effect for the production/service distinction, but with a much lower F value, $F(1, 236) = 8.833; p < .003$, where the mean scores of production organizations were higher than those of service organizations. Reward orientation was the only scale in which there were significant main effects for the production/service dichotomy. There were no significant interaction effects in the perceived Reward Orientation scale.

Hypothesis 2

To test for the hypothesis that policies promoting efficiency will be significantly more prevalent in private compared to public sector organizations, two-way ANOVAs were conducted for each of the efficiency and each of the nonefficiency scales. The efficiency-related scales included reward orientation, improvement of methods and efficiency, task clarity, and task autonomy—sig-

nificance. The nonefficiency scales included participation and openness, and task variety and development.

In support of Hypothesis 2, the private/public sector main effects were significant or highly significant in all efficiency related scales (see Table 3). Private sector means were in all cases higher than those of the public sector. In two of the four scales, however, there were also interaction effects. The reward orientation scale discussed above revealed the most highly significant private/public sector main effects, $F(1, 236) = 25.194, p < .000$. Similarly, the task autonomy and significance scale revealed significant main effects for the private/public sector dichotomy, $F(1, 236) = 8.763, p < .003$, and no interaction effects with production/service. In the remaining two efficiency scales, although main effects for sector of ownership were also significant or nearly significant—improvement of methods and efficiency, $F(1, 236) = 3.856, p < .051$, and task clarity $F(1, 236) = 4.617, p < .033$, the interaction effects were in both cases highly significant: in the improvement of methods and efficiency scale, $F(1, 236) = 11.934, p < .001$, and in the task clarity scale $F(1, 236) = 18.290, p < .000$. Examination of the cell means revealed the same pattern in both scales: The private service cell mean was significantly higher than the other three—private production, public service, and public production (using Scheffé's contrast, $p < .05$; see Figure 1). In both scales the private–public sector differences were greater in service organizations.

The two nonefficiency scales produced no significant main effects for either sector of ownership or type of organization. As seen in Table 3, private/public main effects did not reach significance in either the participation and openness scale, $F(1, 236) = 2.238, p < .136$, or the task variety scale, $F(1, 236) = 0.908, p < .342$.

Hypothesis 2 predicted the differences between private and public sectors to be significant in efficiency-related areas. To provide a further test of this hypothesis, and to determine whether there was indeed a difference between the efficiency and nonefficiency items in terms of the private/public differences, a t test was performed. The test compared the average differences between the means of private and public sectors across the 25 efficiency items, to the average differences across the 13 nonefficiency items. The difference in difference scores was significant at the $p < .05$ level, $t(36) = 2.2$, one-tailed test, providing additional support for the second hypothesis of this study.

Hypothesis 3

To test for the hypothesis that satisfaction would be significantly higher among private sector compared to public sector managers, two-way ANOVAs for the satisfaction responses to the six scales were performed (see Table 4). There were significant main effects for sector of ownership in all six satisfaction scales: Private sector managers reported consistently higher mean levels of satisfaction than the public sector managers. These differences were highly significant ($p < .010$) in five out of the six scales and were significant ($p < .050$) in the sixth scale of participation and openness. There were no significant main effects for the production/service dichotomy in any of the scales. The ANOVAs revealed interaction effects in four scales: reward orientation, $F(1, 236) = 4.424, p < 0.036$, improvement of methods and efficiency, $F(1, 236) = 9.962, p < 0.002$, task clarity, $F(1, 236) = 5.595, p < 0.019$, and task

Table 3

ANOVA for the Effects of Private and Public Sectors and for Production Service Organizations: Existence Assessment

Scale	Item No.	Item content	Sum of source of variation	Squares	df	MS	F	P
Reward Orientation								
			Main effects					
	OC 4	Monetary rewards contingent	Private/Public (PRVPUB)	17.300	1	17.300	25.194	0.000
	JC 12	Monetary rewards according to contribution	Production/Service (PROSER)	6.065	1	6.065	8.833	0.003
	OC 13	Activities for employee welfare	Two-way interactions PRVPUB × PROSER	1.284	1	1.284	1.870	0.173
	OC 5	Performance evaluation and feedback	Explained	24.649	3	8.216	11.966	0.000
	OC 12	Reward for ability and leadership	Residual	162.051	236	0.687		
	OC 14	Emphasis on quality performance	Total	186.700	239	0.781		
	JC 13	Advice for improvement of methods						
Improvement of Methods and Efficiency								
			Main effects					
	OC 9	Communication & information flow	Private/Public (PRVPUB)	3.363	1	3.363	3.856	0.051
	OC 7	Training and efficiency improvement	Production/Service (PROSER)	1.732	1	1.732	1.986	0.160
	OC 15	Improvement of work methods	Two-way interactions PRVPUB × PROSER	10.408	1	10.408	11.934	0.001
	OC 8	Consideration for lower levels	Explained	15.503	3	5.168	5.925	0.001
	OC 6	Improvement of working conditions	Residual	205.823	236	0.872		
			Total	221.326	239	0.926		
Task Autonomy and Significance								
			Main effects					
	JC 9	Independence in decision making	Private/Public (PRVPUB)	6.738	1	6.738	8.763	0.003
	JC 20	Important and significant objectives	Production/Service (PROSER)	0.838	1	0.838	1.090	0.298
	JC 19	Independence of thought and action	Two-way interactions PRVPUB × PROSER	0.047	1	0.047	0.062	0.804
	JC 6	Complete tasks	Explained	7.624	3	2.541	3.305	0.021
	JC 7	Freedom to discuss problems—support	Residual	181.470	236	0.769		
			Total	189.094	239	0.791		
Task Clarity								
			Main effects					
	JC 10	Clear criteria for performance	Private/Public (PRVPUB)	2.763	1	2.763	4.617	0.033
	JC 2	Clear procedures	Production/Service (PROSER)	0.163	1	0.163	0.272	0.602
	JC 17	Clear authority and responsibility	Two-way interactions PRVPUB × PROSER	10.944	1	10.944	18.290	0.000
	JC 18	Feedback on performance	Explained	13.870	3	4.623	7.727	0.000
	JC 1	Precise job definition	Residual	141.210	236	0.598		
	OC 17	Rules and procedures	Total	155.079	239	0.649		
	JC 5	Resources for training						
	JC 3	Required interaction						
Participation and Openness								
			Main effects					
	OC 11	Openness in relations	Private/Public (PRVPUB)	1.302	1	1.302	2.238	0.136
	OC 3	Acceptance and facing of problems	Production/Service (PROSER)	1.202	1	1.202	2.066	0.152
	OC 16	Consideration for preferences	Two-way interactions PRVPUB × PROSER	0.043	1	0.043	0.073	0.787
	OC 18	Freedom & personal responsibility	Explained	2.547	3	0.849	1.459	0.226
	JC 4	Friendship opportunities	Residual	137.325	236	0.582		
	OC 10	Teamwork and cooperation	Total	139.072	239	0.585		
	OC 2	Participative decision making						
	OC 1	Clear organizational goals and objectives						
	OC 19	Flexibility in decision making						

Table 3 (continued)

Scale	Item No.	Item content	Sum of source of variation	Squares	df	MS	F	P
Main effects								
Task Variety and Development	JC 15	Continued skill development	Private/Public (PRVPUB)	0.651	1	0.651	0.908	0.342
	JC 8	Opportunities for interaction	Production/Service (PROSER)	1.837	1	1.837	2.562	0.111
	JC 11	Variety in job activities						
	JC 16	Opportunity for skill utilization						
Two-way interactions								
			PRVPUB × PROSER	2.109	1	2.109	2.941	0.088
			Explained	4.598	3	1.533	2.137	0.096
			Residual	169.271	236	0.717		
			Total	173.869	239	0.727		

Note. JC = Job Characteristics; OC = Organizational Climate.

variety and development, $F(1, 236) = 6.063$, $p < 0.015$. Examination of the cell means revealed that the pattern of interaction effects was very similar in all these four scales and actually identical in all but the reward orientation scale. Figure 2, depicting the mean satisfaction responses to the task clarity scale, represents the pattern of interaction effects in all three scales: Mean levels of satisfaction with task clarity, as well as with improvement of methods and efficiency and task variety, were significantly higher among private service managers than among the other three groups—all managers of the public sector and the private production organizations managers (using Scheffé's contrast $p < .05$; see Figure 2). With respect to reward orientation, public service managers expressed significantly lower mean levels of satisfaction than did the remaining three groups (using Scheffé's contrasts $p < .05$; see Figure 3). The results of all three interaction effects further reinforce the direction of the private/public sector differences in satisfaction. The analysis of contrasts pointed to specific subgroups in the private or public sector that were significantly different, but these variations as well were in the direction of higher satisfaction for private compared to public sector. The private service cell mean was significantly higher than all others in three scales, and the public service cell mean was the lowest in one scale. These findings, along with the significance of the main effects for sector of ownership in all scales, provide strong support for the third hypothesis on significantly higher levels of satisfaction among managers of the private sector. As with the existence assessment, the differences between the two sectors in satisfaction were larger in service organizations.

Discussion

The data of this study contribute to the body of literature on the differences between practices in the private and public sectors by drawing on a sample of top managers from a broad range of Israeli organizations. The findings of this research highlight the similarities to comparable studies in the United States and point to the pervasiveness of differences between private and public sectors across seemingly strict cultural barriers.

The largest differences between the two sectors were in the area of reward systems. The results of this study provided strong support for the hypothesis that rewards of private sector organizations will be contingent on performance to a significantly higher degree than those of public sector organizations. Satisfaction with these policies was also significantly higher among

private sector managers. Conversely, rewards represented the area of greatest dissatisfaction among service organizations of the public sector.

These findings reinforce those of previous studies demonstrating differences in the incentive structures of the two sectors. Research has in the past indicated that money as an incentive is both more readily available and considered as more important among employees of private organizations (Lawler, 1971; Schuster et al., 1973; Rawls et al., 1975). As reported in Lawler's own study, and other cited studies (Lawler, 1971), as well as the research by Rawls et al. (1975), pay is considered to be more important among employees of business organizations than among those in government. Additionally, differences in managerial attitudes relative to both the value of pay and other rewards, and the existence of rewards in the private compared to the public sector emerged in Schuster's (1974) study. Unlike private sector managers, those of the public sector in his study reported that rewards were not distributed based on performance. Finally, Schuster et al. (1973) indicated that a curvilinear relation between amounts of pay and the subjective value attached to pay existed in the private but not the public sector.

The lower satisfaction with rewards in the public sector service organizations in this study is also in line with previous findings of the U.S. Federal Employee Attitude Survey (1980), which explored the attitudes and perceptions of employees towards federal personnel management practices. Dissatisfaction with present pay was expressed by approximately half of the Senior Executive Service group, and this percentage increased for the next lower level of supervisors.

Insight into the causes of dissatisfaction with rewards among employees of government service organizations in the U.S. is also provided from an additional report by the National Commission of Productivity and Work Quality (1975). Merit increases were used by only 42% of the respondents from local governments surveyed. As the report concludes, in some areas "the output-oriented merit increases were in fact longevity increases, and focused more on personal characteristics rather than output" (p. 44). These findings, along with those of the present research, confirm the limited use of performance based rewards and point to the need for redesigning the reward systems of the public sector organizations in order to increase the levels of motivation.

The results of this study provided support for the second hypothesis, predicting that policies that promote efficiency would

Table 4

ANOVA for the Effects of Private and Public Sectors and for Production Service Organizations: Satisfaction Assessment

Scale	Item no.	Item content	Sum of source of variation	Squares	df	MS	F	p
Main effects								
Reward Orientation	OC 4	Monetary rewards contingent	Private/Public (PRVPUB)	19.597	1	19.597	36.182	0.000
	JC 12	Monetary rewards according to contribution	Production/Service (PROSER)	1.352	1	1.352	2.496	0.115
	OC 13	Activities for employee welfare	Two-way interactions PRVPUB × PROSER	2.396	1	2.396	4.424	0.036
	OC 5	Performance evaluation and feedback	Explained	23.345	3	7.782	14.367	0.000
	OC 12	Reward for ability and leadership	Residual	127.821	236	0.542		
	OC 14	Emphasis on quality performance	Total	151.166	239	0.632		
	JC 13	Advice on improvement of methods						
Main effects								
Improvement of Methods and Efficiency	OC 9	Communication & information flow	Private/Public (PRVPUB)	6.472	1	6.472	9.396	0.002
	OC 7	Training and efficiency improvement	Production/Service (PROSER)	1.566	1	1.566	2.274	0.133
	OC 15	Improvement of work methods	Two-way interactions PRVPUB × PROSER	6.861	1	6.861	9.962	0.002
	OC 8	Consideration for lower levels	Explained	14.899	3	4.966	7.211	0.000
	OC 6	Improvement of working conditions	Residual	162.539	236	0.689		
			Total	177.438	239	0.742		
Main effects								
Task Autonomy and Significance	JC 9	Independence in decision making	Private/Public (PRVPUB)	9.766	1	9.766	13.043	0.000
	JC 20	Important and significant objectives	Production/Service (PROSER)	0.769	1	0.769	1.027	0.312
	JC 19	Independence of thought and action	Two-way interactions PRVPUB × PROSER	2.086	1	2.086	2.786	0.096
	JC 6	Complete tasks	Explained	12.621	3	4.207	5.618	0.001
	JC 7	Freedom to discuss problems—support	Residual	176.705	236	0.749		
			Total	189.326	239	0.792		
Main effects								
Task Clarity	JC 10	Clear criteria for performance	Private/Public (PRVPUB)	6.216	1	6.216	11.774	0.000
	JC 2	Clear procedures	Production/Service (PROSER)	1.524	1	1.524	2.887	0.091
	JC 17	Clear authority and responsibility	Two-way interactions PRVPUB × PROSER	2.954	1	2.954	5.595	0.019
	JC 18	Feedback on performance	Explained	10.694	3	3.565	6.752	0.000
	JC 1	Precise job definition	Residual	124.596	236	0.528		
	OC 17	Rules and procedures	Total	135.290	239	0.566		
	JC 5	Resources for training						
	JC 3	Required interaction						
Main effects								
Participation and Openness	OC 11	Openness in relations	Private/Public (PRVPUB)	2.538	1	2.538	5.703	0.018
	OC 3	Acceptance and facing of problems	Production/Service (PROSER)	0.621	1	0.621	1.396	0.239
	OC 16	Consideration for preferences	Two-way interactions PRVPUB × PROSER	0.002	1	0.002	0.004	0.951
	OC 18	Freedom & personal responsibility	Explained	3.161	3	1.054	2.367	0.072
	JC 4	Friendship opportunities	Residual	105.033	236	0.445		
	OC 10	Teamwork and cooperation	Total	108.194	239	0.453		
	OC 2	Participative decision making						
	OC 1	Clear organizational goals and objectives						
	OC 19	Flexibility in decision making						

Table 4 (continued)

Scale	Item no.	Item content	Sum of source of variation	Squares	df	MS	F	p
Main effects								
Task Variety and Development	JC 15	Continued skill development	Private/Public (PRVPUB)	5.032	1	5.032	8.001	0.005
	JC 8	Opportunities for interaction	Production/Service (PROSER)	1.709	1	1.709	2.717	0.101
	JC 11	Variety in job activities	Two-way interactions					
	JC 16	Opportunity for skill utilization	PRVPUB $\times$ PROSER	3.813	1	3.813	6.063	0.015
Explained				10.553	3	3.518	5.594	0.001
Residual				148.408	236	0.629		
Total				158.960	239	0.665		

Note. JC = Job Characteristics; OC = Organizational Climate.

be more prevalent in private sector organizations compared to public sector organizations. Differences were indeed significant in all efficiency-related areas, including improvement of methods, task clarity, task autonomy and significance, and reward orientation. The first two areas were particularly developed in service organizations of the private sector. No differences between the private and public managers in perceived levels of participation or in task variety and development were noted in the study, suggesting that practices that are not directly related to task efficiency do not necessarily differ between the two sectors.

Similar variables to those of this research were found to have the greatest differential effect on the motivation of public sector

employees in a number of U.S. studies. The Federal Employee Attitude Survey (1980) pointed out performance appraisal, feedback, and rewards for performance as the specific areas in which dissatisfaction was expressed in the public sector. The fairness of performance appraisal, and its utility in training programs or improving performance was questioned, whereas a need for more frequent feedback was indicated. Senior level employees expressed dissatisfaction towards the practices in selection, training, rewards, and delegation of authority. Another study by the National Center of Productivity (1978) reported lower technical competence among supervisors of the public sector compared to their private counterparts. These findings might reflect

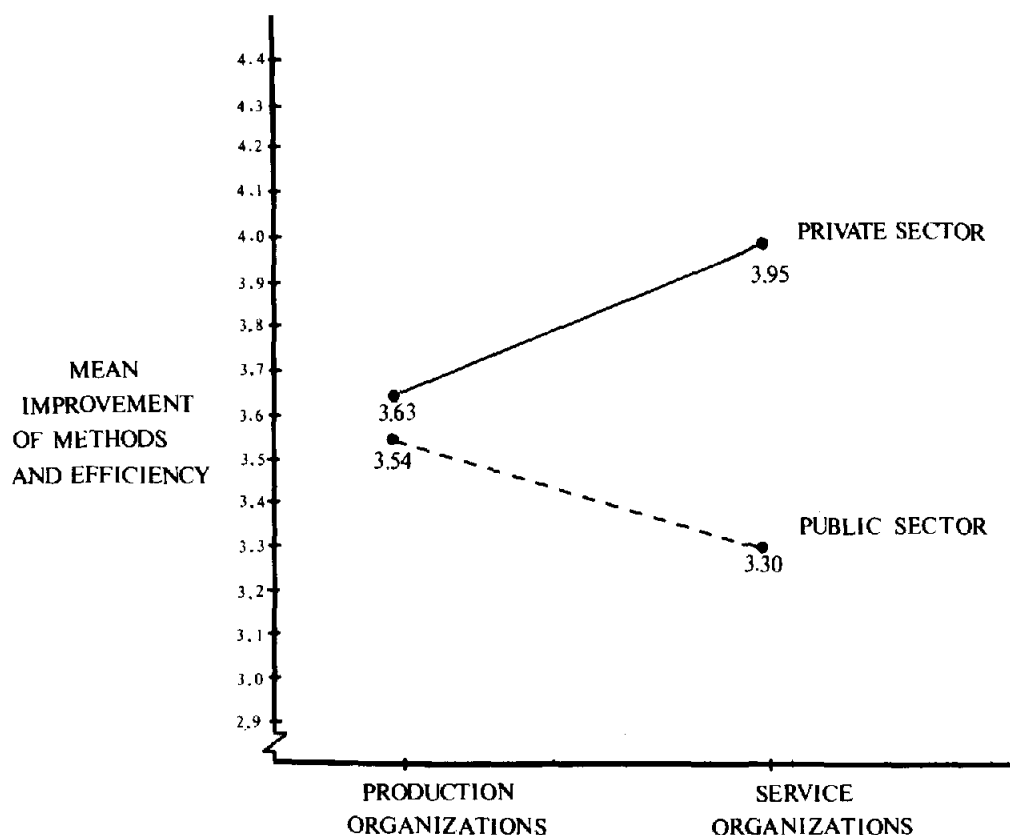


Figure 1. Interaction effects of the private/public sector and production/service organizations: Existence Assessment of the Improvement of Methods and Efficiency Scale.

either lower concern and less investment in training on the part of public sector organizations, or inability to recruit quality personnel. Top level personnel in the Federal Employee Attitude Survey (1980) reported having an insufficient degree of authority to hire and reward competent individuals, and need was expressed for civil service reform to increase perceptions that appointments to senior positions are merit based.

Private sector managers in this research reported significantly higher satisfaction than the public sector, providing support for the third hypothesis as well. Because subsamples of both sectors consisted of production and service organizations, an analysis of those variations revealed that in specific areas, differences in satisfaction were more amplified in the service organizations. This was true of reward systems, improvement of methods and efficiency, task clarity, and task variety and development.

A growing number of studies in other cultural settings, mainly the United States, indicate similar results of consistently lower levels of satisfaction in the public sector. Paine et al. (1966) found need satisfaction across all areas studied to be lower among managers in a governmental agency compared to a similar group from industry. Porter and Mitchell (1967) reported that commissioned military officers were less fulfilled and less satisfied than similar levels of civilian managers, and Rhinehart et al. (1969) determined that managers of a Veterans Administration

agency were less satisfied than a comparison group from business and industry. Buchanan (1974) found first-year job challenge among industrial managers to be significantly greater than their public counterparts, and Hayward (1978) suggested, as a result of his research on job satisfaction in the public sector, that business organizations might provide baselines against which government employees can be compared. Finally, Rainey's (1979) results are consistent with this trend, in reporting that government managers scored lower on scales of satisfaction with promotion and with co-workers than did business managers.

These findings permit some generalizations about the unique dynamics and constraints of the public sector (Perry & Porter, 1982) that dictate less efficient policies. Policy makers in the public sector are faced with special organizational imperatives. They confront controversial political and often value-laden issues very specific to their society. The size of many governmental organizations, the pluralism in policy implementation networks, and the inadequately developed performance criteria, render the task of channelling sanctioned goal-directed activities more difficult. The kind of formal bureaucracies that characterize most public organizations have usually emphasized reward structures that embody security and tenure, and are founded on political considerations rather than concerns with efficiency.

Von Mises (1944), describing the European society in his pe-

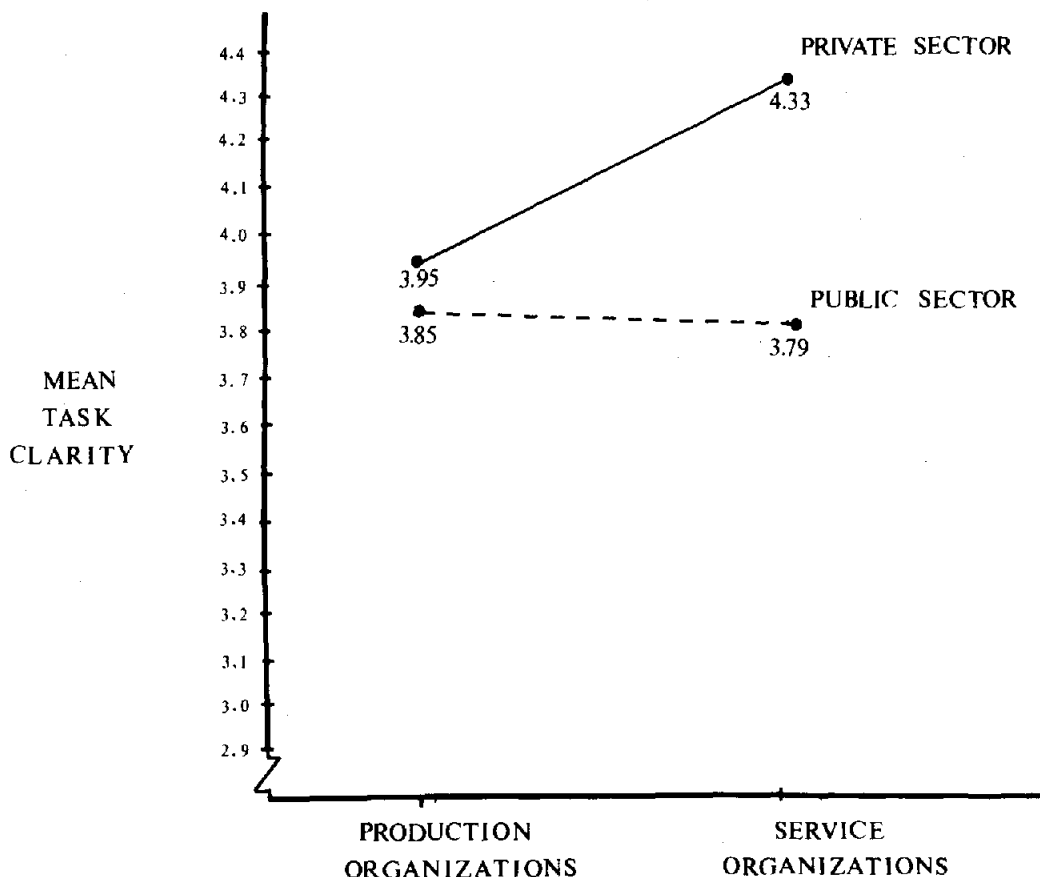


Figure 2. Interaction effects of the private/public sector and production/service organizations: Satisfaction Assessment of the Task Clarity Scale.

riod in the classic, *Bureaucracy*, viewed public administration as the handling of the government apparatus of coercion and compulsion. He concluded that no reform can remove the formalistic and bureaucratic features of the governmental structure. This study's findings suggest that the functioning of bureaucracy, no matter what cultural framework it exists in, will be recognizably similar and subject to similar constraints.

Implications of these findings point to the need for restructuring reward systems and applying goal setting and job redesign-enrichment methods as means of increasing satisfaction in the public sector. The potential success of goal setting, however, is constrained by the vagueness and conflicting nature of public sector goals (Perry & Porter, 1982). Furthermore, the lower levels of task autonomy and significance that have also been documented by most other studies in this area can be attributed to the additional constraints imposed on public managers by legislative and accountability requirements. Because autonomy and self-direction are considered to be basic motivating properties central to redesign-enrichment efforts, and because in the case of public managers these factors are at least in part externally controlled, application of such motivating techniques have to be specifically modified for the public sector and can only be applied within specific limitations.

Conclusions

In generalizing from the results of this study and drawing implications for the United States, some attention should be paid to cross-cultural differences, which cannot be underestimated. This study consists of a survey of private and public sector managers in Israel. Both sectors in the study included service and production organizations. Public industries in Israel are owned by the Histadrut, the federation of workers. There is no equivalent group in the United States, because there are no nationalized industries in this country. It is important to note that there are differences between the United States, European countries, and Israel that might have implications for assessing cross-cultural attitudes of the managers tested within their framework of traditional values. Unlike countries such as France, Sweden, West Germany, and Israel, where the notion of distributive justice as well as socialistic principles are part of the culture, in the United States, a typically individualistic country, the ideologies of meritocracy and Darwinism are prevalent. Because the public sector is viewed to be a critical and indivisible part of the system in Israel, we might expect some of the differences between the private and public sectors to be obscured. Strikingly, this was not the case in our study. In addition to variations in historical and socio-

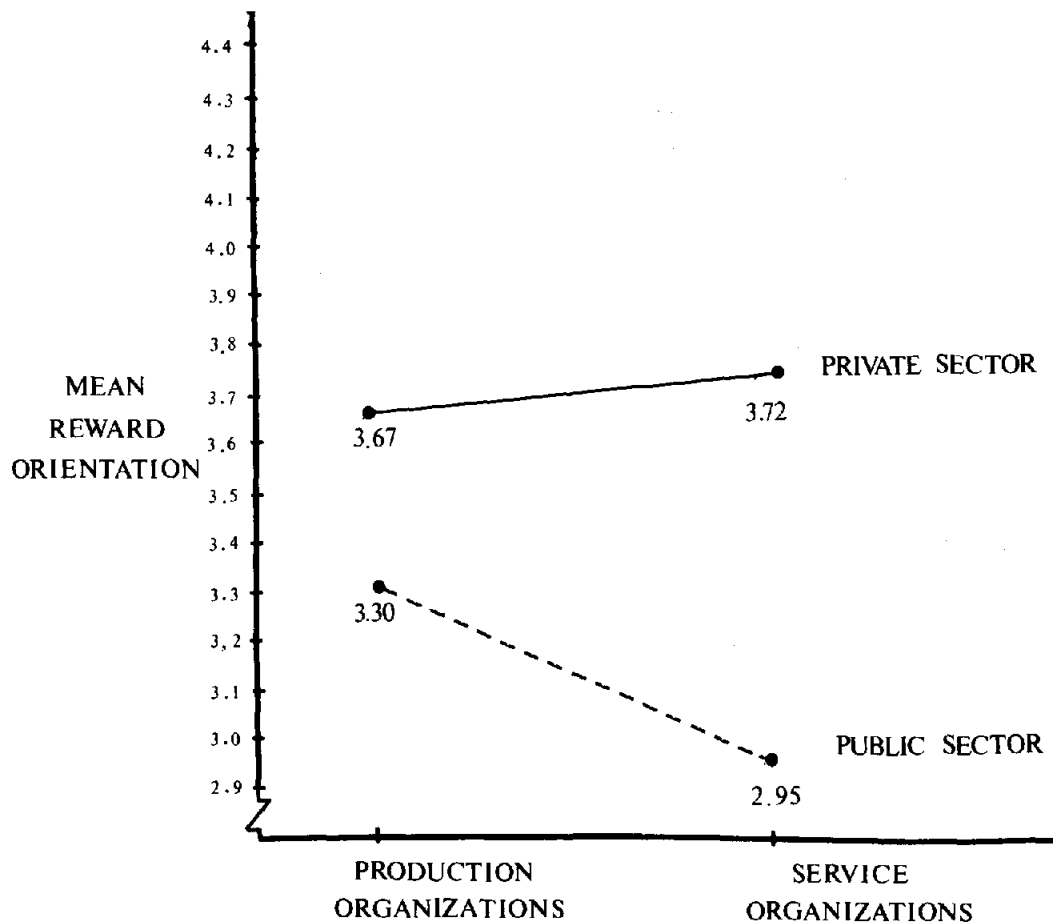


Figure 3. Interaction effects of the private/public sector and production/service organizations: Satisfaction Assessment of the Reward Orientation Scale.

political backgrounds, further key differences between Israel and the United States in terms of size, the scale of the economic system, and the pressures pushing up the Israeli rate of inflation should be mentioned. All of these have appropriate implications for translation of the meaning of rewards. Still, although the generalizability of the data presented here is constrained by these limitations, the findings are very similar to those appearing in the literature on the United States, highlighting the significance of the private-public sector differences across cultures.

This study offers some new insights on the accumulated evidence on the dynamics of the public and private sectors, and some directions for future research. Differences in policies between the two sectors had a more critical and measurably different impact in service rather than on production organizations, as reported by the managers of this study. There are no comparable findings in the United States, because no public sector group of industries is equivalent to the Histadrut-owned industries. It would, however, be useful for future research in the United States and in other countries to differentiate systematically between production and service private sector organizations when comparing them to public sector organizations. Such studies would provide further tests of the present findings and might offer important new insights on the dynamics of the differences between the private and public sectors.

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